Orb cup: Option A vs Option B

Uniquea · Derived inside the model (reduced one-variable Orb cup, Master §3.15) · run November 5, 2026 · from Orb-Option-A-vs-B.xlsx

> **The point of this and the point of that; the complete answer is between those two points.** Option A keeps the Orb's history in where it tips; Option B gives nearly one answer and keeps the history in the Alt Field. Between them runs one dial, γT.

What each option yields

Measure	Option A (γT = 0)	Option B (γT = 10)	Note
Largest shift	6.339%	-0.041%	MAX over the three starting wobbles
Smallest shift	0.982%	-0.047%	MIN over the three starting wobbles
Spread factor (largest ÷ smallest size)	6.46	1.14	A: history shows in where it tips. B: nearly one answer
Mean shift	3.241%	-0.044%	B's mean ≈ the contraction's lag past the fold
Max ledger residual (absolute)	1.9×10^{-16}	1.6×10^{-16}	Books balance to machine precision
Where the memory lives	In the Orb (tipping point)	In the Alt Field ($E_Alt - E_0$ nearly constant)	Return address on the box
New parameters needed	None	Damping γ	The real case is set by γT, still Open

All twelve runs

Wobble E_0	γT	Option	Measured shift (m_c−m)/m_c	Predicted shift_A·e^(−0.8γT)	Measured ÷ predicted	E_Alt	E_Alt − E_0	Ledger residual
1×10^{-9}	0	A	6.339%	6.339%	1.00	0	—	-1.9×10^{-16}
1×10^{-9}	1	Between	2.901%	2.848%	1.02	5.340×10^{-10}	-4.66×10^{-10}	-6.3×10^{-18}
1×10^{-9}	3	Between	0.555%	0.575%	0.97	8.860×10^{-10}	-1.14×10^{-10}	-2.5×10^{-16}
1×10^{-9}	10	B	-0.041%	0.002%	-19.41	1.060×10^{-9}	6.04×10^{-11}	6.7×10^{-17}
3×10^{-10}	0	A	2.404%	2.404%	1.00	0	—	-1.1×10^{-16}
3×10^{-10}	1	Between	1.048%	1.080%	0.97	1.679×10^{-10}	-1.32×10^{-10}	-4.3×10^{-17}
3×10^{-10}	3	Between	0.192%	0.218%	0.88	2.845×10^{-10}	-1.55×10^{-11}	-8.2×10^{-18}
3×10^{-10}	10	B	-0.044%	0.001%	-54.35	3.796×10^{-10}	7.96×10^{-11}	-1.6×10^{-16}
1×10^{-10}	0	A	0.982%	0.982%	1.00	0	—	-1.3×10^{-16}
1×10^{-10}	1	Between	0.403%	0.441%	0.91	6.198×10^{-11}	-3.80×10^{-11}	2.1×10^{-17}

1×10^{-10}	3	Between	0.046%	0.089%	0.51	1.124×10^{-10}	1.24×10^{-11}	-9.4×10^{-17}
1×10^{-10}	10	B	-0.047%	0.000%	-143.37	1.850×10^{-10}	8.50×10^{-11}	1.5×10^{-16}

Blue rows: Option A. Orange rows: Option B. Positive shift = tips before the fold; negative = carried just past it by the contraction's lag. The prediction law holds while the shift is well above the lag floor (ratios 0.97–1.02 at γT = 1) and fails at γT = 10, where the lag (≈ −0.045%) sets the tipping point.

Model inputs

Quantity	Value	Note
$\bar{A}_2$ (cup stiffness scale)	1	Units of $\bar{A}_2$; set to 1 in the run
ε_L (Lila gap, as used)	0.004344	Lila gap ε_L ≈ 0.434% (Master §3.12)
$\bar{A}_4 = -(\bar{A}_2/6)(1 - \varepsilon_L)$	-0.165942667	Lila mapping (Master §3.15)
$\bar{T}_1 = 2\bar{A}_2 + 4\bar{A}_4$	1.336229333	Perch pinned at y = 1 with no triad
Edge distance $\delta = \bar{A}_4 + \bar{A}_2/6$	0.000724	Distance from the tap edge
Barrier at rest ≈ $\varepsilon^3/3 \cdot \bar{A}_2$ (first order)	2.732×10^{-8}	Simulation gave 2.750×10^{-8} (exact)
Fold point m_c ($I \propto \chi^2$)	4.7658×10^{-6}	From root-finding in adiabatic.py (Nov 5)
Contraction time T	1×10^{6}	Ramp of m from 0 to m_c over T
Wobble period	67.41	From cup curvature (adiabatic.py)
Wobbles during contraction	14,835	
Damping law exponent (predicted)	0.8	From J^(4/5) scaling: shift_B ≈ shift_A·e^(−0.8·γT)

Shelves, caveats and credits

- Shelf: Derived inside the model (reduced one-variable Orb cup, Master §3.15). Not a test against data.
- Option A: no damping; the wobble's action J is carried, and the tipping point forms a family X_*(E_i(0)).
- Option B: weak damping γ; the wobble relaxes and its energy is booked to the Alt Field, giving nearly one tipping point.
- Between: shift_B ≈ shift_A·e^(−0.8·γT). The true case depends on γT (wobble fade time vs contraction time): Open.
- Ledger: E_now − E_start − work by contraction + E_Alt = 0 to ~1e-16 in all twelve runs (item 3, for this reduced model).
- Caveat: a faster contraction carries small wobbles past the fold (known lag effect). Runs used T = 1e6 (~14,800 wobbles).
- Still Open (Phantom's list): full state vector; coupled equations with a living a(t); flux-freezing case; late-time test in the full model.
- Owner's principle: correct answers lie between two points; when answers aren't in the box, look for the return address on the outside.
- Phantom (Humor): 'Faulty theory and equations unravel very quickly like a loose piston rod … KABOOM, Free Fireworks Show.' HAT: the part is the connecting rod; Top Fuel engines run near 8,000 rpm.
- Credits: adiabatic invariants (Einstein, Solvay 1911; Ehrenfest); slow passage through a fold (bifurcation delay) is standard dynamical-systems theory.
- Files: Orb-Adiabatic-Check/optionAB.py, ab.json, adiabatic.py.